Junye Zhou

Portfolio (moonlitjaderabbit.github.io) | junye.zhou@mail.utoronto.ca

Education

University of Toronto Toronto, ON

Bachelor of Applied Science in Computer Engineering *Expected Graduation: 2029 + PEY*

- Coursework: Algorithms & Data Structures (Python, C), Programming Fundamentals (C++), Digital Systems (Verilog), Electrical Circuits, Intro to Electronics, Linear Algebra.

Experience

Drone Team Member Sep 2026 – Present

RSX Robotics for Space Exploration *Toronto, ON*

- Used KiCad and performed soldering for ongoing drone power distribution board and radio-control link projects, supporting the electrical subteam's work on onboard power and communication systems.
- Review UART connections and SiK/MAVLink telemetry for the Kakute H7 flight controller. Support the planned **2.4 GHz** to **900 MHz** ExpressLRS control-link transition to separate the drone's control frequency from the rover's existing **2.4 GHz** system.

Engineering Design Team Feb 2026 – Apr 2026

Variable Resistance Swim Training System | Praxis II *University of Toronto*

- Designed a capstan-based swim-training concept for the Mississauga Aquatic Club, targeting **8 resistance levels (60–170 N)** to provide adjustable training loads.
- Compared resistance concepts using Pugh matrices, considering adjustability, manufacturability, and poolside safety in the team's concept selection.
- Wrote engineering requirements and verification plans to define how the concept would be assessed; presented the team's design at the Praxis II Showcase.

Projects

Trench–Hydro One Innovation Sprint - Winning Concept Oct 2026

Team Lead | **5th Place** | *Partial-Discharge Detection* | *Python, openEMS*

- Led the team's development of a combined electromagnetic/acoustic concept for detecting partial discharge in capacitor voltage transformers.
- Simulated **300–700 MHz** electromagnetic pulse propagation with Python and openEMS in a simplified capacitor voltage transformer model. Represented partial discharge with a simulated source inside the oil and examined predicted electromagnetic fields at the housing.
- Compared noisy receiver waveforms and detection thresholds; documented source and material assumptions, model simplifications, and outstanding validation needs for the combined EM/acoustic concept.

Matboard Bridge Design Challenge - Winning Design Nov 2025 – Jan 2026

Bridge Design Team | **2nd Place** | *CIV102*

- Designed, analyzed, and fabricated a **1.25 m** matboard bridge with the team, earning **1st in the cohort** and **2nd in the EngSci class**.
- Cross-checked team-developed Python structural analysis against hand calculations to verify predicted loads and failure modes, supporting the team's design decisions.

Toronto Badminton Drop-in Finder Website Personal Project

https://torontodropin.ca/

- Built a website using City of Toronto open data to help badminton players find nearby community centres and keep track of drop-in sessions.
- Included filters for sport, day, and location, nearby-centre sorting, and saved favourites so users can narrow their options and revisit preferred centres.

Technical Skills

Programming: C, C++, Python, Java, Verilog
Tools: Quartus Prime, ModelSim, DESim, KiCad, LTspice, MATLAB, OSP Tracker, openEMS, Cursor
Engineering: Requirements definition, Pugh matrices, concept evaluation, verification planning

For project details and additional work, visit my portfolio: https://moonlitjaderabbit.github.io/